GOD: Enhancing Generalization via Deep Grafting for Sequential Recommendation

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Abstract

Sequential recommenders often struggle with sparse and noisy histories, limiting generalization to unseen interactions. Knowledge distillation mitigates this by transferring dense supervision from a teacher to a student. However, most distillation methods run teacher and student independently, then match student outputs or representations to the teacher. Such supervision entangles student-component effects, blurring whether weak generalization stems from unreliable embeddings, overfitted encoding, or co-adaptation to sparse histories. In this paper, we propose **G**raft-**O**riented **D**istillation (**GOD**), a component-level distillation framework for improved generalization through grafting. Grafting denotes replacing selected frozen-teacher components with trainable student counterparts to build hybrid source models. GOD uses these hybrid models to evaluate student embeddings with the teacher encoder and the student encoder with teacher embeddings, providing component-level feedback. At inference, GOD uses only the student, incurring no additional cost. Across three real-world datasets, GOD outperforms state-of-the-art baselines by up to 13.92%.

CCS Concepts

• **Information systems** → **Recommender systems**; **Personalization**; **Collaborative filtering**.

Keywords

Sequential Recommendation, Knowledge Distillation, Grafting

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1 Introduction

Sequential recommendation (SR) aims to predict the next item a user will interact with by modeling sequential patterns in interaction histories [22, 46, 50, 59]. Deep sequential models [12, 25, 29, 45, 48], particularly Transformer-based recommenders [10, 17, 26, 28, 42, 63], have achieved strong performance by capturing complex dependencies among past interactions. However, real-world user histories are often sparse and noisy [30, 51, 68], making SR models prone to fitting observed co-occurrences or spurious transitions rather than reliable patterns for future interactions [31, 62]. Therefore, a central challenge in SR is learning robust sequential representations that generalize beyond limited and noisy training histories [6, 8, 38, 60].

Knowledge distillation (KD) [13, 39] can alleviate data sparsity by transferring dense supervision from a pretrained teacher to a compact student. Existing recommendation KD methods have explored diverse forms of supervision, including ranking outputs [15, 23, 47], embeddings [2, 16], interest-level knowledge [9], and self-distilled signals [8, 27, 57, 64]. Despite their effectiveness, most methods use the teacher primarily as an external signal generator. The teacher first produces predictions or representations through its own inference path, and the student is then trained to match these signals through a separate inference path.

KD with separated inference paths is particularly limiting in SR since sparsity and noise affect the components that form sequence representations. Most SR models map items into embeddings and transform the resulting sequence with an encoder [12, 17, 69]. Sparse and noisy interactions can yield unreliable student embeddings, while the encoder can amplify spurious dependencies in short histories. In a student-only path, embedding and encoder errors can be entangled, making it difficult to provide component-specific guidance for generalization. Thus, KD can better support generalization through component-level feedback beyond matching teacher-produced signals from separately executed paths.

Intermediate-representation KD [14, 39] can partially mitigate this issue, yet it still compares targets from separate teacher and student paths. Such feedback blurs whether weak generalization stems from unreliable embeddings, limited encoding, or their co-adaptation to sparse histories. A more direct strategy is to replace a single teacher component with its student counterpart while keeping the rest teacher-side. The resulting hybrid path tests student

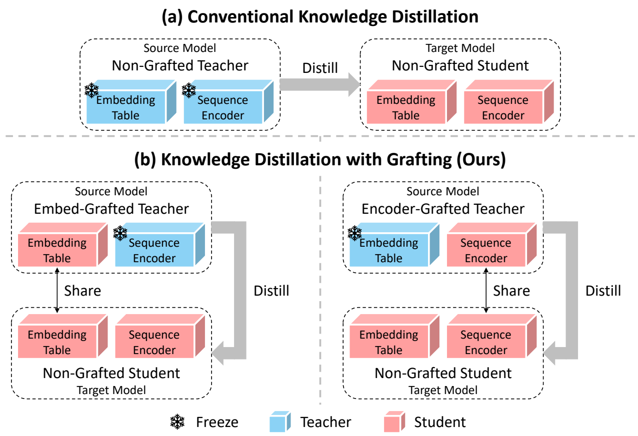


Figure 1: Illustration of (a) conventional KD with an independent teacher and (b) KD with grafting. In (b), hybrid source models replace selected teacher components with student counterparts. Inference uses only the non-grafted student.

embeddings under the teacher encoder and the student encoder under teacher embeddings, yielding component-level feedback for enhanced generalization. This aligns with model *grafting*, which combines components from different networks within a single computation path [33, 35, 41]. As illustrated in Figure 1, we use grafting in KD to isolate and supervise student components during training.

Based on grafting, we propose **G**raft-**O**riented **D**istillation (**GOD**), a component-level KD framework for SR. GOD constructs hybrid source models by grafting the student embedding table or the student sequence encoder into a frozen teacher, while the non-grafted student serves as the target model. These grafted sources expose different student components to teacher-side computation, providing complementary supervision for generalization. For Transformer-based SR models, *Grafted Encoding* stabilizes representation generation by enabling mutual attention between teacher- and student-side tokens. GOD then distills the representations generated by the source models with *Graft-aware Contrastive Learning*, which adaptively balances correlated grafted views. During inference, GOD uses only the non-grafted student, adding no extra inference cost.

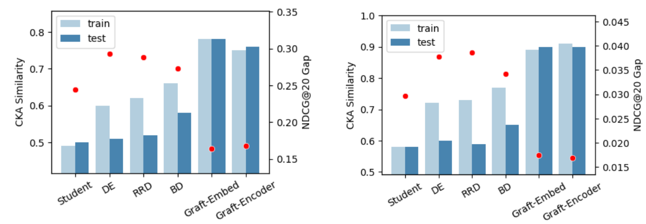
Our main contributions are summarized as follows:

- We revisit KD for SR from a component-level perspective, highlighting how separated teacher-student paths can weaken generalization under sparse histories.
- We propose GOD, a component-level KD framework that constructs hybrid source models by grafting trainable student components into a frozen teacher for enhanced generalization.
- Experiments on three real-world datasets demonstrate that GOD consistently outperforms existing KD and self-supervised baselines by up to 13.92%.

2 Preliminary

2.1 Problem Formulation

In SR, the goal is to predict the next item from the historical interaction sequence of a user. Let $\mathcal{U}$ and $\mathcal{I}$ denote the user and item



(a) Amazon Beauty **(b) MovieLens 1M**

Figure 2: Train/test CKA similarity of teacher-student sequence representations and NDCG@20 gaps with SASRec.

sets, respectively. Each user $u \in \mathcal{U}$ has a chronologically ordered sequence $s_u = [i_{u,1}, i_{u,2}, \ldots, i_{u,|s_u|}]$, where $i_{u,t} \in \mathcal{I}$ is the t-th item interacted with by user u. Given s_u, the prediction is formulated as:

$$\underset{i \in \mathcal{I}}{\operatorname{argmax}}\ p(i_{u,|s_u|+1} = i | s_u). \tag{1}$$

2.2 Sequential Recommender

Most sequential recommenders [12, 17, 46, 69] consist of two components: an embedding table and a sequence encoder. Among them, Transformer-based models [3, 30, 37, 55, 66, 68] have become particularly prevalent due to their effectiveness in modeling complex sequential patterns [7, 49]. We therefore describe the standard Transformer-based formulation used in SR.

2.2.1 Embedding Table. Two learnable embedding tables are used: an item table $I \in \mathbb{R}^{|\mathcal{I}| \times d}$ and a position table $P \in \mathbb{R}^{N \times d}$, where N is the maximum sequence length. After truncating the earliest items or front-padding zeros to length N, s_u is embedded as:

$$e_u = [I_{i_{u,1}} + P_1, I_{i_{u,2}} + P_2, \ldots, I_{i_{u,N}} + P_N] \in \mathbb{R}^{N \times d}, \tag{2}$$

where $I_{i_{u,t}}$ and P_t are the embeddings of item $i_{u,t}$ and position t.

2.2.2 Sequence Encoder. The sequence embedding e_u is processed by an L-layer multi-head Transformer encoder, where $\text{Trm}^l(\cdot)$ denotes the l-th layer. With $H_u^0 = e_u$, each layer outputs:

$$H_u^l = [h_{u,1}^l, h_{u,2}^l, \ldots, h_{u,N}^l] = \text{Trm}^l(H_u^{l-1}) \in \mathbb{R}^{N \times d}. \tag{3}$$

The final-layer last-position representation $h_{u,N}^L$ defines the sequence representation $h_u = f(e_u)$, where $f(\cdot)$ applies all Transformer layers and selects the last position.

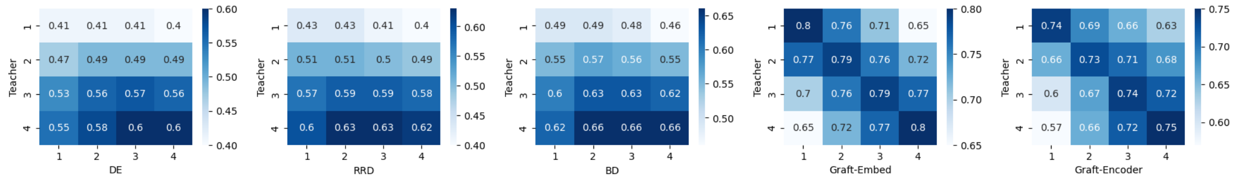
2.3 Next Item Prediction

Given s_u and ground-truth item index g_u, the next-item probability distribution and recommendation loss are computed as:

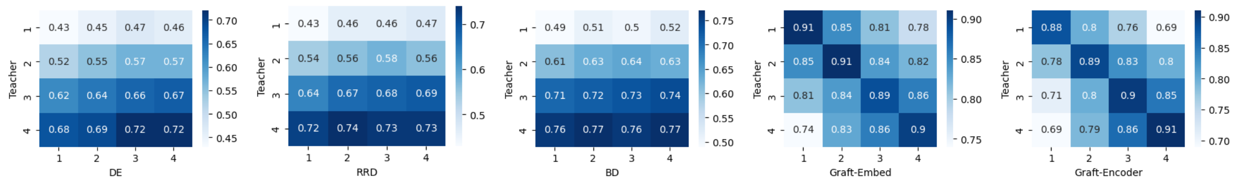
$$\begin{aligned} \hat{y}_u &= \text{Softmax}(h_u I^\top) \in \mathbb{R}^{|\mathcal{I}|}, \\ \mathcal{L}_{rec}(u) &= -\log \hat{y}_u[g_u]. \end{aligned} \tag{4}$$

3 Grafting Analysis for Distillation

Before presenting GOD, we analyze whether grafting can provide component-level feedback for generalization. We evaluate whether grafting improves the transfer of distilled knowledge to unseen sequences and encourages procedural alignment between teacher and student encoding processes.



(a) Amazon Beauty



(b) MovieLens 1M

Figure 3: Layer-wise CKA heatmaps of teacher-student intermediate representations with 4-layer SASRec.

3.1 Generalization of Distilled Knowledge

Grafting improves generalization through component-level feedback. Conventional KD supervises the full student path with signals from an independent teacher, jointly optimizing student embeddings and encoder. Grafting instead replaces a frozen teacher component with its trainable student counterpart, forming a hybrid source model that evaluates the student component under teacher-side computation. Since the grafted component is shared with the student, KD regularization on the student also refines the source model [13, 40], creating an adaptive supervision loop. This resembles bidirectional KD [21, 67] but requires no explicit teacher updates and provides finer-grained guidance than output-level imitation.

Empirical Evidence. We compare Student (no KD) and five KD variants (DE [15], RRD [15], BD [21], Graft-Embed, and Graft-Encoder[1]) using SASRec [17] as the backbone. Figure 2 reports centered kernel alignment (CKA) [20] between teacher and student representations on train/test sequences, along with NDCG@20 gaps (train minus test). Student shows low CKA without KD, while DE and RRD improve train alignment but suffer test-alignment drops and large NDCG@20 gaps. BD partially reduces these gaps through dynamic distillation, but Graft-Embed and Graft-Encoder maintain high train/test CKA with much smaller NDCG@20 gaps, indicating better generalization of distilled knowledge.

3.2 Procedural Alignment

Beyond generalization, we analyze procedural alignment, i.e., layer-wise correspondence between teacher and student encoding processes. Conventional KD supervises the student with teacher signals from separate paths, which can encourage output imitation rather than aligned encoding behavior [41]. Distilling intermediate representations [14, 39] or attention maps [70] may expose internal teacher states, but these targets are still produced by separate teacher forward passes. Grafting instead places trainable student components inside teacher-conditioned paths, guiding student encoding without explicitly matching every internal teacher state.

[1]Graft-Embed and Graft-Encoder use the *Embed-Grafted Teacher* and *Encoder-Grafted Teacher* in Figure 1 as their source models, respectively.

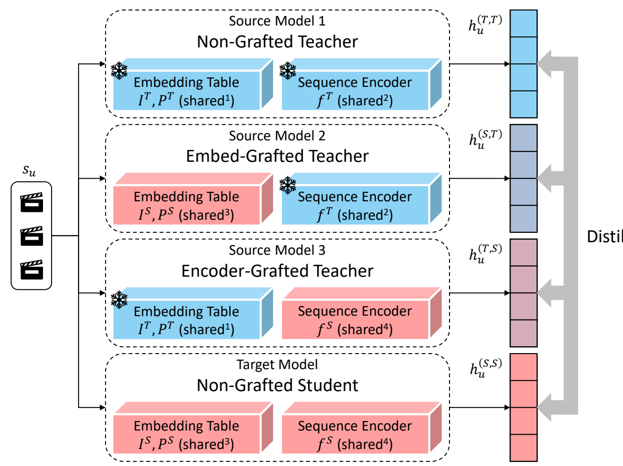


Figure 4: Illustration of GOD. GOD distills representations of three source models—*Non-Grafted Teacher*, *Embed-Grafted Teacher*, and *Encoder-Grafted Teacher*—into *Non-Grafted Student*. Identical "shared" indices indicate parameter sharing.

Empirical Evidence. Figure 3 visualizes CKA heatmaps of teacher–student intermediate representations on training sequences with 4-layer SASRec. DE, RRD, and BD show teacher-final-layer dominance, with all student layers most aligned to the final teacher layer, suggesting output imitation rather than layer-wise procedural alignment. In contrast, Graft-Embed and Graft-Encoder show clear diagonal patterns, with student layers most aligned to corresponding teacher layers. Notably, this correspondence emerges without explicit intermediate representation distillation, suggesting that grafting aligns student and teacher encoding processes through structural coupling rather than direct state matching.

4 Proposed Framework: GOD

We propose **GOD**, a KD framework using hybrid source models constructed via grafting, as illustrated in Figure 4. Each hybrid model replaces the teacher embedding table or sequence encoder with its student counterpart, exposing the student component to teacher-side computation. GOD generates coupled representations with *Grafted Encoding* and distills them with *Graft-aware Contrastive Learning*. At inference, GOD uses only the non-grafted student.

4.1 Source / Target Model Configuration

Following the grafting analysis in Sec. 3, GOD constructs three source models for component-level distillation: *Non-Grafted Teacher*, *Embed-Grafted Teacher*, and *Encoder-Grafted Teacher*. *Non-Grafted Teacher* preserves the original teacher as a stable knowledge source. *Embed-Grafted Teacher* replaces the teacher embedding table with the student embedding table and feeds student embeddings to the teacher encoder. *Encoder-Grafted Teacher* replaces the teacher encoder with the student encoder and processes teacher embeddings. The target model, *Non-Grafted Student*, receives distilled sequence representations from all source models. As shown in Figure 4, all models share frozen-teacher and trainable-student components, introducing no standalone source-model parameters.

Forwarding Process. For a sequence s_u, we first construct teacher- and student-side embeddings. Let I^T, P^T and I^S, P^S denote the item and position embedding tables of the teacher and student, respectively. The embeddings are given by:

$$\begin{aligned} e_u^T &= [I^T_{i_{u,1}} + P^T_1, I^T_{i_{u,2}} + P^T_2, \ldots, I^T_{i_{u,N}} + P^T_N] \in \mathbb{R}^{N \times d^T}, \\ e_u^S &= [I^S_{i_{u,1}} + P^S_1, I^S_{i_{u,2}} + P^S_2, \ldots, I^S_{i_{u,N}} + P^S_N] \in \mathbb{R}^{N \times d^S}. \end{aligned} \tag{5}$$

With $f^T(\cdot)$ and $f^S(\cdot)$ denoting the teacher and student sequence encoders, the two embeddings are paired with the two encoders to produce four representations:

$$\begin{aligned} h_u^{(T,T)} &= f^T(e_u^T) \cdot W_{down} && \in \mathbb{R}^{d^S}, \\ h_u^{(S,T)} &= f^T(e_u^S \cdot W_{up}) \cdot W_{down} && \in \mathbb{R}^{d^S}, \\ h_u^{(T,S)} &= f^S(e_u^T \cdot W_{down}) && \in \mathbb{R}^{d^S}, \\ h_u^{(S,S)} &= f^S(e_u^S) && \in \mathbb{R}^{d^S}, \end{aligned} \tag{6}$$

where $h_u^{(M_1,M_2)}$ denotes the representation generated using the embeddings of M_1 and the sequence encoder of M_2 for $M_1, M_2 \in \{T, S\}$. Thus, $h_u^{(T,T)}$, $h_u^{(S,T)}$, $h_u^{(T,S)}$, and $h_u^{(S,S)}$ are generated by *Non-Grafted Teacher*, *Embed-Grafted Teacher*, *Encoder-Grafted Teacher*, and *Non-Grafted Student*, respectively. GOD uses shared linear projections $W_{down} \in \mathbb{R}^{d^T \times d^S}$ and $W_{up} \in \mathbb{R}^{d^S \times d^T}$ to align dimensions while preserving the structural effect of grafting (see Figure 10 in Sec. 5.4).

4.2 Grafted Encoding

GOD is applicable to SR models with separable embeddings and encoders, while Grafted Encoding (GE) targets Transformer-based models, which dominate recent SR research [3, 30, 37, 55, 66, 68]. GE stabilizes representation generation by concatenating teacher- and student-side embeddings and enabling mutual attention, as illustrated in Figure 5. Early in training, grafted student components are not yet well optimized, which can make hybrid source model representations unstable. Mutual attention lets teacher-side embeddings provide stable context early on, while improved student-side

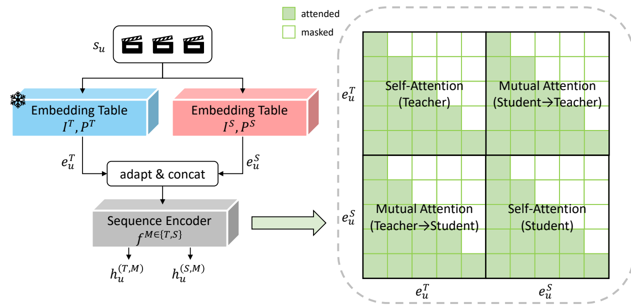


Figure 5: Illustration of Grafted Encoding with teacher-student mutual attention.

embeddings later refine teacher-conditioned representations without updating the frozen teacher [21, 67]. Thus, GE strengthens structural coupling while keeping the teacher frozen.

Forwarding Process. In GE, the four models are grouped by their shared sequence encoder: *Non-Grafted Teacher* and *Embed-Grafted Teacher* share $f^T(\cdot)$, while *Encoder-Grafted Teacher* and *Non-Grafted Student* share $f^S(\cdot)$. For each pair, teacher- and student-side embeddings are adapted to the same dimension, concatenated, and passed through the shared encoder. Let $\mathrm{Trm}^{M,l}(\cdot)$ denote the l-th Transformer layer of encoder $f^{M\in\{T,S\}}$, whose depth and hidden dimension are L^M and d^M, respectively. With $H_u^{T,0} = [e_u^T; e_u^S \cdot W_{up}]$ and $H_u^{S,0} = [e_u^T \cdot W_{down}; e_u^S]$, the forwarding process is:

$$\begin{aligned} H_u^{M,l} &= [h^{M,l}_{u,1}, h^{M,l}_{u,2}, \ldots, h^{M,l}_{u,N}, h^{M,l}_{u,N+1}, \ldots, h^{M,l}_{u,2N}] \\ &= \mathrm{Trm}^{M,l}(H_u^{M,l-1}) \in \mathbb{R}^{2N \times d^M}. \end{aligned} \tag{7}$$

Instead of using separate causal attention for each side, GE enables mutual attention between teacher- and student-side tokens. The causal attention mask is:

$$(A_{causal})_{ij} = \mathbb{I}[j \le i], \quad A_{causal} \in \{0,1\}^{N \times N}, \tag{8}$$

where $(A_{causal})_{ij} = 1$ allows the i-th query token to attend to the j-th key token. For the concatenated sequence, GE uses:

$$A_{GE} = \begin{bmatrix} A_{causal} & A_{causal} \\ A_{causal} & A_{causal} \end{bmatrix} \in \{0,1\}^{2N \times 2N}. \tag{9}$$

Finally, the four sequence representations are obtained from the last valid positions of the teacher- and student-side token blocks:

$$\begin{aligned} h_u^{(T,T)} &= h^{T,L^T}_{u,N} \cdot W_{down} && \in \mathbb{R}^{d^S}, \\ h_u^{(S,T)} &= h^{T,L^T}_{u,2N} \cdot W_{down} && \in \mathbb{R}^{d^S}, \\ h_u^{(T,S)} &= h^{S,L^S}_{u,N} && \in \mathbb{R}^{d^S}, \\ h_u^{(S,S)} &= h^{S,L^S}_{u,2N} && \in \mathbb{R}^{d^S}. \end{aligned} \tag{10}$$

Complexity. Although GE forms a length-$2N$ concatenated sequence, its overhead is limited to training-time attention since GE is not used during inference. Without GE, four length-N paths incur cost $O(2N^2d^T + 2N^2d^S)$, whereas GE uses two length-$2N$ paths with cost $O(4N^2d^T + 4N^2d^S)$, roughly doubling attention computation. This overhead comes from stronger structural coupling and can be offset by faster convergence in practice (see Table 7 in Sec. 5.5). For memory cost, we also evaluate GE with each side sequence halved, so the concatenated sequence keeps length N. GE still improves performance in this setting (see Table 5 in Sec. 5.4).

4.3 Graft-aware Contrastive Learning

Leveraging the four representations, GOD performs KD through contrastive learning. Rather than point-wise matching or logit imitation, contrastive learning transfers relational structure among sequences across views [34, 60]. This is well suited to GOD since representations of source and target models form structurally related views through shared teacher and student components.

GOD considers pairwise relations among all four representations [4, 36, 55]. We include both source-target and source-source pairs because hybrid source models contain trainable student components, allowing source-source relations to regularize the student embeddings or encoder. With $\mathcal{A} = \{(T,T), (S,T), (T,S), (S,S)\}$, the contrastive loss for each pair is:

$$\mathcal{L}_{pair}(h_u^a, h_u^b) = -\log \frac{\exp(h_u^a \cdot h_u^b/\tau)}{\sum_{s_{u'} \in B} \exp(h_u^a \cdot h_{u'}^b/\tau)}, \tag{11}$$

where τ is the temperature and B is the mini-batch of sequences.

The six pair types, however, are not equally informative since the four representations share teacher and student components. Following prior works on similarity-aware pair weighting [18, 44, 53], GOD down-weights highly similar relations to reduce redundant supervision and emphasize complementary component-level signals. Specifically, the pair weight and distillation objective are defined as:

$$\begin{aligned} w(a,b) &= \frac{\exp(\mathrm{sg}(-\frac{1}{|B|}\sum_{s_u \in B} h_u^a \cdot h_u^b))}{\sum_{\{a',b'\} \subset \mathcal{A}} \exp(\mathrm{sg}(-\frac{1}{|B|}\sum_{s_u \in B} h_u^{a'} \cdot h_u^{b'}))}, \\ \mathcal{L}_{kd}(u) &= \sum_{\{a,b\} \subset \mathcal{A}} w(a,b) \mathcal{L}_{pair}(h_u^a, h_u^b), \end{aligned} \tag{12}$$

where $\mathrm{sg}(\cdot)$ denotes stop-gradient. GCL down-weights highly similar pair types and emphasizes less-aligned ones, while stop-gradient keeps $w(a,b)$ as an adaptive coefficient rather than a directly optimized target.

4.4 Inference and Optimization

Inference. GOD uses only *Non-Grafted Student* for inference:

$$p(i_{u,|s_u|+1}|s_u) = \hat{y}_u = \mathrm{Softmax}(h_u I^{S^\top}), \tag{13}$$

where $h_u = f^S(e_u^S)$. Note that with GE, h_u differs from $h_u^{(S,S)}$, which is generated with teacher-student mutual attention for training.
Optimization. Next-item prediction uses the sequence representation of *Non-Grafted Student*, and KD improves its consistency and generalization. Therefore, we jointly optimize the recommendation and distillation objectives with loss coefficient λ:

$$\mathcal{L} = \sum_{u \in \mathcal{U}} (\mathcal{L}_{rec}(u) + \lambda \mathcal{L}_{kd}(u)). \tag{14}$$

5 Experiment

In this section, we address the following research questions (RQs):

- **RQ1**: How does GOD perform compared with existing KD and self-supervised methods in SR? (Section 5.2)
- **RQ2**: Does GOD improve generalization and robustness under challenging conditions? (Section 5.3)
- **RQ3**: How effective are the core components of GOD? (Section 5.4)
- **RQ4**: How efficient and hyperparameter-sensitive is GOD in practice? (Section 5.5)

Table 1: Statistics of the datasets.

Datasets	# Interactions	# Users	# Items	Avg.SeqLen	Sparsity
Amazon Beauty	198,502	22,363	12,101	8.9	99.93 %
Yelp	345,186	29,915	21,572	11.5	99.95 %
MovieLens 1M	999,611	6,040	3,416	165.5	95.16 %

5.1 Experimental Setup

5.1.1 Dataset. To validate the effectiveness of GOD, we conduct experiments on three public datasets: Amazon Beauty [32], Yelp [1], and MovieLens 1M [11]. These datasets vary in domain, scale, and sparsity, with detailed statistics provided in Table 1. Following [17, 38], we treat interactions as implicit feedback, and filter out users and items with fewer than five interactions.

5.1.2 Baseline. We first describe the backbone models to which GOD and all KD methods are applied. We then summarize the KD baselines used for comparison.

- **Backbone Recommender**: We apply GOD and all KD methods to GRU4Rec [12], FMLPRec [69], and SASRec [17], covering RNN-, MLP-, and Transformer-based architectures. For each backbone, Student is the compact model trained without KD, and Teacher is the larger pretrained model. GE is applied only to SASRec.
- **General Recommendation KD**: We compare with KD methods for general recommendation. RD [47] transfers teacher ranking knowledge. CD [23] samples informative items from teacher rankings. DE [15] distills latent embeddings. RRD [15] relaxes teacher rankings for supervision. HTD [16] transfers hierarchical topology knowledge in the representation space. BD [21] performs bidirectional distillation for recommendation.
- **SR-tailored KD**: We further compare with KD designed for SR. AdaRec [2] searches for scene-adaptive student architectures. MSKDIK [9] distills interest representation and drift knowledge in multiple stages. EMKD [8] uses ensemble modeling with contrastive distillation.

We exclude LLM-based KD methods because they use external LLM knowledge or language-model students, while our experiments focus on ID-based KD with the same interaction-only input and backbone recommenders.[2] This ensures that performance differences reflect the distillation strategy rather than additional modality or model-scale advantages.

5.1.3 Evaluation. For evaluation, we adopt the leave-one-out protocol [17], where the last item of each chronological sequence is used for testing, the penultimate item for validation, and the rest for training. We perform full-ranking without negative sampling [38]. Performance is measured by HR@k and NDCG@k, where $k \in \{10, 20\}$.

[2] We discuss their relationship to GOD in Section 6.1.

Table 2: Overall performance. Best and second-best results are marked in bold and underlined. *Improv.S* and *Improv.B* denote relative improvements over Student and the strongest baseline. Asterisk (*) indicates statistical significance at $p < 0.05$ by paired t-test over five independent runs against the strongest baseline.

Dataset	Backbone	Metric	Teacher	Student	RD	CD	DE	RRD	HTD	BD	AdaRec	MSKDIK	EMKD	GOD	*Improv.S*	*Improv.B*
Amazon Beauty	GRU4Rec	HR@10	0.0506	0.0384	0.0392	0.0403	0.0413	0.0426	0.0438	0.0464	0.0450	0.0454	<u>0.0472</u>	**0.0508***	32.29 %	7.63 %
		HR@20	0.0788	0.0622	0.0626	0.0639	0.0652	0.0668	0.0682	0.0711	0.0692	0.0703	<u>0.0722</u>	**0.0777***	24.92 %	7.62 %
		NDCG@10	0.0257	0.0187	0.0190	0.0196	0.0203	0.0210	0.0217	0.0232	0.0224	0.0227	<u>0.0237</u>	**0.0257***	37.69 %	8.44 %
		NDCG@20	0.0322	0.0249	0.0262	0.0266	0.0270	0.0275	0.0281	0.0290	0.0285	0.0286	<u>0.0294</u>	**0.0316***	26.91 %	7.48 %
	FMLPRec	HR@10	0.0667	0.0524	0.0540	0.0547	0.0555	0.0564	0.0573	0.0583	0.0583	0.0590	<u>0.0596</u>	**0.0634***	20.99 %	6.38 %
		HR@20	0.0959	0.0813	0.0843	0.0849	0.0856	0.0864	0.0871	0.0883	0.0873	0.0886	<u>0.0892</u>	**0.0951***	16.97 %	6.61 %
		NDCG@10	0.0368	0.0296	0.0315	0.0319	0.0322	0.0326	0.0330	0.0336	0.0335	0.0338	<u>0.0341</u>	**0.0372***	25.68 %	9.09 %
		NDCG@20	0.0436	0.0346	0.0363	0.0368	0.0372	0.0377	0.0382	0.0388	0.0388	0.0393	<u>0.0396</u>	**0.0427***	23.41 %	7.83 %
	SASRec	HR@10	0.0727	0.0605	0.0617	0.0626	0.0635	0.0646	0.0655	0.0675	0.0666	0.0668	<u>0.0683</u>	**0.0725***	19.83 %	6.15 %
		HR@20	0.1030	0.0925	0.0932	0.0943	0.0955	0.0969	0.0981	0.1006	0.0995	0.0999	<u>0.1017</u>	**0.1051***	13.62 %	3.34 %
		NDCG@10	0.0401	0.0316	0.0321	0.0327	0.0332	0.0339	0.0346	0.0360	0.0353	0.0355	<u>0.0364</u>	**0.0392***	24.05 %	7.69 %
		NDCG@20	0.0472	0.0390	0.0410	0.0415	0.0420	0.0426	0.0432	0.0444	0.0437	0.0439	<u>0.0447</u>	**0.0474***	21.54 %	6.04 %
Yelp	GRU4Rec	HR@10	0.0498	0.0365	0.0383	0.0379	0.0389	0.0391	0.0395	0.0403	0.0399	0.0404	<u>0.0407</u>	**0.0456***	24.93 %	12.04 %
		HR@20	0.0843	0.0652	0.0686	0.0675	0.0701	0.0712	0.0725	0.0742	0.0739	0.0751	<u>0.0761</u>	**0.0810***	24.23 %	6.44 %
		NDCG@10	0.0250	0.0187	0.0195	0.0191	0.0197	0.0201	0.0204	0.0208	0.0206	0.0210	<u>0.0212</u>	**0.0241***	28.88 %	13.68 %
		NDCG@20	0.0337	0.0245	0.0263	0.0257	0.0266	0.0272	0.0277	0.0283	0.0278	0.0286	<u>0.0290</u>	**0.0322***	31.43 %	11.03 %
	FMLPRec	HR@10	0.0543	0.0407	0.0421	0.0429	0.0438	0.0443	0.0449	0.0460	0.0454	0.0456	<u>0.0464</u>	**0.0508***	24.82 %	9.48 %
		HR@20	0.0873	0.0647	0.0676	0.0684	0.0700	0.0708	0.0715	0.0729	0.0723	0.0724	<u>0.0735</u>	**0.0804***	24.27 %	9.39 %
		NDCG@10	0.0293	0.0223	0.0226	0.0231	0.0239	0.0244	0.0249	0.0258	0.0253	0.0255	<u>0.0262</u>	**0.0287***	28.70 %	9.54 %
		NDCG@20	0.0375	0.0282	0.0296	0.0301	0.0311	0.0314	0.0316	0.0322	0.0320	0.0320	<u>0.0324</u>	**0.0360***	27.66 %	11.11 %
	SASRec	HR@10	0.0568	0.0454	0.0481	0.0470	0.0485	0.0490	0.0494	0.0503	0.0497	0.0500	<u>0.0507</u>	**0.0567***	24.89 %	11.83 %
		HR@20	0.0917	0.0764	0.0808	0.0791	0.0814	0.0818	0.0821	0.0828	0.0824	0.0826	<u>0.0831</u>	**0.0929***	21.60 %	11.79 %
		NDCG@10	0.0303	0.0241	0.0259	0.0249	0.0263	0.0265	0.0267	0.0271	0.0269	0.0270	<u>0.0273</u>	**0.0311***	29.05 %	13.92 %
		NDCG@20	0.0390	0.0316	0.0344	0.0334	0.0352	0.0356	0.0361	0.0370	0.0364	0.0367	<u>0.0373</u>	**0.0405***	28.16 %	8.58 %
MovieLens 1M	GRU4Rec	HR@10	0.2063	0.1325	0.1344	0.1334	0.1353	0.1361	0.1369	0.1375	0.1380	0.1386	<u>0.1391</u>	**0.1499***	13.13 %	7.76 %
		HR@20	0.2976	0.1975	0.2030	0.1999	0.2046	0.2061	0.2076	0.2086	0.2096	0.2105	<u>0.2116</u>	**0.2231***	12.96 %	5.43 %
		NDCG@10	0.1012	0.0665	0.0683	0.0671	0.0684	0.0686	0.0688	0.0689	0.0691	0.0690	<u>0.0692</u>	**0.0770***	15.79 %	11.27 %
		NDCG@20	0.1260	0.0867	0.0895	0.0873	0.0909	0.0917	0.0924	0.0930	0.0934	0.0940	<u>0.0945</u>	**0.1029***	18.69 %	8.89 %
	FMLPRec	HR@10	0.1996	0.1338	0.1354	0.1349	0.1361	0.1363	0.1367	0.1369	0.1371	0.1373	<u>0.1375</u>	**0.1514***	13.15 %	10.11 %
		HR@20	0.2894	0.2123	0.2167	0.2133	0.2200	0.2215	0.2231	0.2242	0.2253	0.2264	<u>0.2274</u>	**0.2398***	12.95 %	5.45 %
		NDCG@10	0.0962	0.0694	0.0700	0.0696	0.0719	0.0729	0.0738	0.0745	0.0751	0.0757	<u>0.0764</u>	**0.0833***	20.03 %	9.03 %
		NDCG@20	0.1205	0.0841	0.0884	0.0852	0.0890	0.0894	0.0897	0.0899	0.0901	0.0903	<u>0.0905</u>	**0.0992***	17.95 %	9.61 %
	SASRec	HR@10	0.2330	0.1640	0.1664	0.1654	0.1683	0.1695	0.1707	0.1715	0.1723	0.1731	<u>0.1739</u>	**0.1856***	13.17 %	6.73 %
		HR@20	0.3301	0.2490	0.2542	0.2513	0.2579	0.2598	0.2617	0.2630	0.2643	0.2655	<u>0.2668</u>	**0.2813***	12.97 %	5.43 %
		NDCG@10	0.1150	0.0816	0.0839	0.0820	0.0856	0.0865	0.0873	0.0879	0.0885	0.0890	<u>0.0896</u>	**0.0978***	19.85 %	9.15 %
		NDCG@20	0.1417	0.1052	0.1086	0.1058	0.1090	0.1091	0.1093	0.1094	0.1096	0.1097	<u>0.1098</u>	**0.1218***	15.78 %	10.93 %

5.1.4 Implementation. We implement all models in PyTorch on a single NVIDIA GeForce RTX 3090 GPU and report averages over five independent runs. Unless otherwise specified, the default student and teacher dimensions are $d^S = 16$ and $d^T = 64$, respectively. For all backbones, we use 2 encoder layers, a dropout rate of 0.5, and a maximum sequence length $N = 50$. For SASRec, we use 2 attention heads. For fair comparison, we thoroughly tune each baseline following the search ranges in its original paper. For GOD, the KD loss coefficient λ is selected from {1e-4, 1e-3, 1e-2, 1e-1, 1e-0}, and the temperature τ from {1, 5, 10, 20}. We train all models with Adam optimizer [19] using a learning rate of 0.001 and batch size of 256. Early stopping is applied with patience 30 based on validation NDCG@20.

5.2 Overall Performance (RQ1)

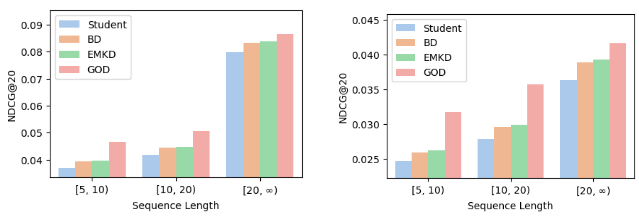
Table 2 reports the KD performance of GOD in SR. Overall, GOD achieves the best results across all datasets, backbones, and metrics, improving over the strongest baseline by up to 13.92%. Beyond these gains, we make three key observations:

- All KD methods outperform Student, confirming that teacher supervision helps compact models under sparse sequential feedback. SR-tailored KD methods generally outperform general recommendation KD methods, highlighting the importance of sequence-aware distillation.
- BD is a strong general KD baseline. On the sparser Amazon Beauty and Yelp datasets, BD often even surpasses AdaRec and MSKDIK, suggesting that adaptive teacher-student supervision benefits sparse and evolving interactions. However, unlike BD, GOD keeps the teacher frozen while still achieving better performance.
- GOD shows larger gains over Student on Amazon Beauty and Yelp than on the denser MovieLens 1M. This supports our motivation that grafting improves the generalization of distilled knowledge under sparse interactions, where SR models are prone to overfitting [31, 62]. Consistent gains across GRU4Rec, FMLPRec, and SASRec further show the broad applicability of GOD across architectures.

Self-Distillation. Table 3 reports performance with SASRec when teacher and student have the same capacity ($d^S = d^T = 64$). For GOD and KD baselines, this setting uses an offline pretrained teacher with the same architecture and dimension as the student, allowing us to examine whether GOD remains effective without a capacity gap. Since no model compression is involved, we also

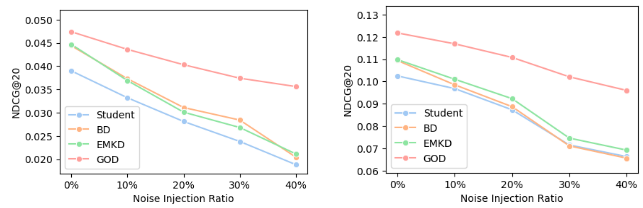
Table 3: Self-distillation performance with SASRec. Best and second-best results are marked in bold and underlined. *Improv.* denotes the relative improvement over the strongest baseline. Asterisk (*) indicates statistical significance at $p < 0.05$ by paired t-test over five independent runs against the strongest baseline.

Dataset	Metric	SASRec	RRD	HTD	BD	MSKDIK	EMKD	CL4SRec	MCLRec	CT4Rec	DuoRec	RCL	GOD	*Improv.*
Amazon Beauty	HR@10	0.0727	0.0718	0.0721	0.0749	0.0763	0.0785	0.0796	0.0806	0.0837	<u>0.0856</u>	0.0853	**0.0923***	7.83 %
	HR@20	0.1030	0.1010	0.1018	0.1047	0.1062	0.1078	0.1084	0.1099	0.1139	0.1164	<u>0.1172</u>	**0.1249***	6.57 %
	NDCG@10	0.0401	0.0396	0.0400	0.0426	0.0438	0.0454	0.0464	0.0474	0.0490	0.0506	<u>0.0517</u>	**0.0559***	8.12 %
	NDCG@20	0.0472	0.0461	0.0469	0.0499	0.0513	0.0525	0.0544	0.0554	0.0568	0.0580	<u>0.0596</u>	**0.0641***	7.55 %
Yelp	HR@10	0.0568	0.0559	0.0571	0.0574	0.0566	0.0584	0.0588	0.0591	0.0601	0.0606	<u>0.0607</u>	**0.0640***	5.44 %
	HR@20	0.0917	0.0903	0.0931	0.0934	0.0917	0.0945	0.0951	0.0957	0.0968	0.0975	<u>0.0978</u>	**0.1032***	5.52 %
	NDCG@10	0.0303	0.0296	0.0311	0.0311	0.0304	0.0304	0.0305	0.0306	0.0310	<u>0.0311</u>	0.0310	**0.0329***	5.79 %
	NDCG@20	0.0390	0.0382	0.0393	0.0397	0.0385	0.0391	0.0390	0.0396	0.0401	<u>0.0404</u>	0.0403	**0.0427***	5.69 %
MovieLens 1M	HR@10	0.2330	0.2266	0.2303	0.2318	0.2271	0.2336	0.2345	0.2304	<u>0.2352</u>	0.2347	0.2308	**0.2461***	4.63 %
	HR@20	0.3301	0.3236	0.3288	0.3285	0.3243	0.3317	0.3348	0.3275	0.3370	<u>0.3382</u>	0.3342	**0.3554***	5.09 %
	NDCG@10	0.1150	0.1133	0.1147	0.1155	0.1141	0.1177	0.1201	0.1188	0.1214	0.1230	<u>0.1232</u>	**0.1295***	5.11 %
	NDCG@20	0.1417	0.1390	0.1408	0.1419	0.1393	0.1442	0.1482	0.1455	0.1490	0.1497	<u>0.1500</u>	**0.1561***	4.07 %



(a) Amazon Beauty **(b) Yelp**

Figure 6: Performance by sequence length with SASRec.



(a) Amazon Beauty **(b) MovieLens 1M**

Figure 7: Performance under different noise injection ratios with SASRec.

compare with self-supervised contrastive baselines that use online signals generated by the model itself during training, such as augmented or consistency-based views: CL4SRec [60], MCLRec [36], CT4Rec [4], DuoRec [38], and RCL [55]. GOD achieves the best results across all datasets and metrics, improving over the strongest baseline by up to 8.12%. We make three observations:

- Conventional KD methods provide limited gains in this same-capacity setting. Several methods improve only marginally or even underperform SASRec. This suggests that, without a capacity gap, simply imitating a same-capacity teacher provides limited additional knowledge for recommendation.
- Self-supervised methods are stronger than KD baselines. This indicates that representation regularization is more effective than direct teacher imitation in the self-distillation setting. However, these methods rely on augmentation- or consistency-based views rather than teacher-conditioned source views.
- GOD consistently outperforms both KD and self-supervised baselines. This shows that its gains do not merely come from model compression or generic contrastive learning. Instead, grafted teacher-student views provide more transferable supervision to improve same-capacity sequence representation learning.

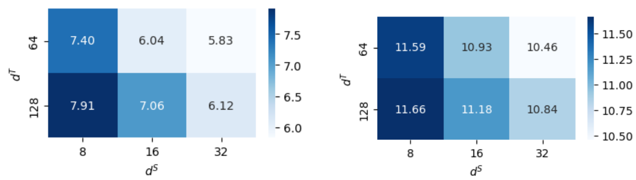
5.3 Generalization and Robustness (RQ2)

To further understand the generalization behavior of GOD, we evaluate its robustness under both data-side and teacher-side challenges. We consider user history sparsity and interaction noise as data-side factors, and teacher-student capacity gap and teacher quality as teacher-side factors.

User History Sparsity. We group users by sequence length and evaluate each group separately. As shown in Figure 6, all methods improve with longer sequences, indicating that longer histories provide richer preference signals. More importantly, GOD consistently performs best across all groups, with especially large margins in short-sequence groups. This confirms that GOD is particularly effective when user interactions are sparse, where conventional KD methods have limited evidence for learning robust sequential patterns. The margin becomes smaller for long-sequence users, suggesting that the benefit of grafting is most pronounced when generalization from sparse interactions is needed.

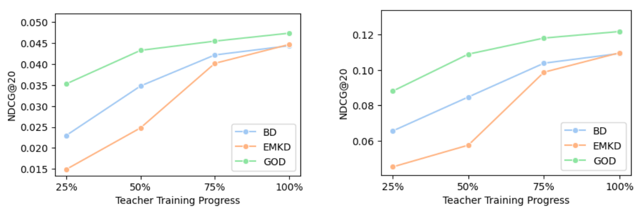
Noise Robustness. We perturb the input sequence at test time by replacing a given ratio of items with random negative items, where the ratio is measured relative to the sequence length. As shown in Figure 7, performance consistently drops as the noise ratio increases, confirming that noisy interactions make sequential pattern modeling more challenging. Nevertheless, GOD maintains the best performance across all noise levels, and its performance gap remains clear even under high noise ratios. This indicates that grafting provides more robust component-level supervision than bi-directional KD or SR-tailored KD, allowing the student to preserve transferable sequential patterns when observed histories are corrupted.

Capacity Sensitivity. We vary the student dimension $d^S \in \{8, 16, 32\}$ and teacher dimension $d^T \in \{64, 128\}$ to examine robustness of GOD across teacher-student capacity gaps. As shown in Figure 8,



(a) Amazon Beauty (b) MovieLens 1M

Figure 8: Relative NDCG@20 improvements (%) of GOD over EMKD, the strongest baseline, under different student and teacher dimensions with SASRec.



(a) Amazon Beauty (b) MovieLens 1M

Figure 9: Performance under different teacher training progress with SASRec. Here, 100% denotes the best teacher checkpoint.

GOD consistently improves over EMKD across all dimension pairs. The gains are larger when the student is smaller, indicating that grafting is especially useful when compact students lack sufficient capacity to absorb teacher knowledge. In addition, increasing the teacher dimension generally brings larger improvements, suggesting that GOD can better exploit richer teacher knowledge instead of suffering from a wider teacher-student gap. These results show that GOD remains effective across diverse compression ratios.

Teacher Quality Sensitivity. We evaluate sensitivity to teacher quality by freezing checkpoints at 25%, 50%, 75%, and 100% of the training progress toward the best teacher checkpoint. For example, if the best teacher checkpoint is obtained at epoch 44, the 50% checkpoint is trained for 22 epochs. As shown in Figure 9, all methods benefit from more trained teachers, confirming that teacher quality affects KD performance. Notably, BD outperforms EMKD with weaker teachers, suggesting that bidirectional updates can partially compensate for weak teacher supervision. GOD, however, consistently performs best without explicitly updating the teacher. This indicates that grafting and GE provide adaptive teacher-conditioned supervision efficiently, enabling GOD to extract transferable knowledge even from partially trained teachers.

5.4 Ablation Study (RQ3)

To understand the contribution of each design in GOD, we conduct ablation studies on three core components: source models, GE, and GCL.

Table 4: Source model ablation with SASRec. NDCG@20 results are shown for the source models used in KD, where (T,T), (S,T), and (T,S) denote *Non-Grafted Teacher*, *Embed-Grafted Teacher*, and *Encoder-Grafted Teacher*, respectively. Case (1) corresponds to GOD.

Case	(T,T)	(S,T)	(T,S)	Amazon Beauty	Yelp	MovieLens 1M
(1)	○	○	○	**0.0474**	**0.0405**	**0.1218**
(2)	○	○		0.0451	0.0375	0.1141
(3)	○		○	0.0448	0.0376	0.1135
(4)		○	○	0.0410	0.0342	0.1084
(5)	○			0.0421	0.0349	0.1111
(6)		○		0.0401	0.0336	0.1074
(7)			○	0.0399	0.0337	0.1071

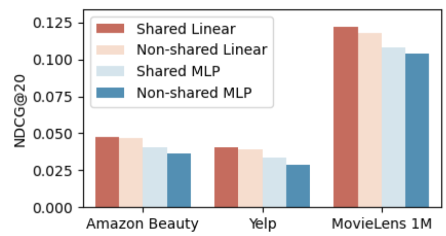


Figure 10: Projection design ablation with SASRec. Non-shared uses source model-specific projections, and MLP uses 2-layer projections. Shared Linear corresponds to GOD.

Source Models. Table 4 analyzes the contribution of each source model. Case (1), corresponding to full GOD, achieves the best performance on all datasets. Removing either hybrid source in cases (2) and (3) consistently degrades performance, confirming that *Embed-Grafted Teacher* and *Encoder-Grafted Teacher* provide complementary structural coupling. The larger drop in case (4) shows that *Non-Grafted Teacher* remains important as a stable knowledge anchor. Cases (5), (6), and (7) further show that any single source alone is insufficient, indicating that GOD benefits from jointly distilling stable teacher knowledge and complementary grafted views.

We further compare the shared linear projections (i.e., W_{down} and W_{up}) used in GOD with non-shared linear projections and 2-layer MLP projections. As shown in Figure 10, the shared linear design consistently achieves the best performance across all datasets. Non-shared linear projections and MLP-based projections underperform despite their higher flexibility, indicating that the gains of GOD do not come from additional projection capacity. This supports our design choice of using simple shared projections, which adapt dimensions while preserving the structural effect of grafting.

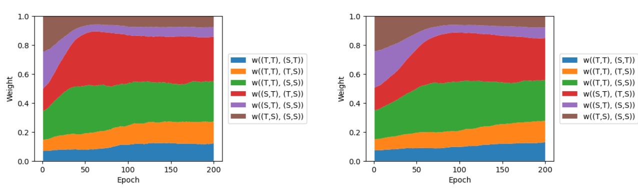
Grafted Encoding. Table 5 evaluates GE variants, with full GE achieving the best performance on all datasets. Uni-directional variants, i.e., T→S only and S→T only, outperform w/o GE, showing that cross-side token attention is useful even in one direction. Among them, T→S only performs better than S→T only, suggesting that teacher-side tokens effectively stabilize student-side representations. Notably, GE (half length) outperforms w/o GE, although its concatenated sequence length matches the original input length.

Table 5: GE ablation with SASRec. NDCG@20 is shown by varying teacher-student token interaction. T→S only allows student-side query tokens to attend to teacher-side key/value tokens, while S→T only allows the reverse direction. GE (half length) halves each side sequence length so that the concatenated sequence has the original length.

Variant	Amazon Beauty	Yelp	MovieLens 1M
GE	**0.0474**	**0.0405**	**0.1218**
T→S only	0.0466	0.0394	0.1184
S→T only	0.0459	0.0386	0.1166
GE (half length)	0.0462	0.0396	0.1152
w/o GE	0.0452	0.0379	0.1126

Table 6: GCL ablation with SASRec. NDCG@20 is shown by varying pair construction and weighting. Source-Target only uses three pairs involving $h_u^{(S,S)}$, Unweighted uses $\mathcal{L}_{pair}$ instead of $\mathcal{L}_{kd}$, and Learnable weight replaces similarity-based weights with trainable pair weights.

Variant	Amazon Beauty	Yelp	MovieLens 1M
GCL	**0.0474**	**0.0405**	**0.1218**
Source-Target only	0.0460	0.0386	0.1111
Unweighted	0.0455	0.0379	0.1090
Learnable weight	0.0427	0.0359	0.1074



(a) Amazon Beauty **(b) MovieLens 1M**

Figure 11: Dynamics of six GCL pair weights over training. Higher pair similarity yields a lower weight.

This shows that GE does not merely benefit from a longer attention span. Finally, all partial variants remain inferior to full GE, indicating that bidirectional mutual attention with full-length histories is necessary to fully exploit teacher-student coupling.

Graft-aware Contrastive Learning. Table 6 analyzes the effect of GCL. Full GCL achieves the best performance on all datasets, showing the benefit of adaptive pair-type weighting. Source-Target only performs better than Unweighted, indicating that adding source-source pairs without proper weighting can introduce redundant or imbalanced supervision. However, GCL further improves over Source-Target only, confirming that source-source pairs are useful when their contributions are dynamically controlled. Learnable weight performs worst, suggesting that freely learnable pair weights are unstable and that similarity-based weighting with stop-gradient provides a more reliable distillation budget allocation.

To further understand GCL, we visualize the normalized weights of all six pairwise contrastive terms over training. As shown in Figure 11, the weights do not collapse to a single pair but remain distributed across multiple relations, confirming that GCL performs balanced reweighting. Early in training, pairs involving $h_u^{(S,S)}$ with grafted sources, i.e., $w((S,T),(S,S))$ and $w((T,S),(S,S))$, receive relatively high weights, while their importance gradually decreases as the student becomes better aligned with the sources. In contrast, $w((T,T),(S,S))$ and $w((S,T),(T,S))$ gain higher weights and remain important in later epochs, indicating that GCL reallocates the distillation budget toward more informative and less redundant relations.

Table 7: Efficiency analysis with SASRec. Mem. denotes peak GPU memory, Time/Epoch denotes average training time per epoch, and Train denotes wall-clock training time until the best validation checkpoint.

Model	Amazon Beauty				MovieLens 1M			
	Mem.	Time/Epoch	Train	NDCG@20	Mem.	Time/Epoch	Train	NDCG@20
Teacher	0.7 GB	8.35 s	6.12 m	0.0472	1.1 GB	15.88 s	23.82 m	0.1417
Student	0.6 GB	7.75 s	24.03 m	0.0390	0.8 GB	15.13 s	18.66 m	0.1052
BD	1.0 GB	12.98 s	27.69 m	0.0444	1.5 GB	26.67 s	59.12 m	0.1094
EMKD	1.1 GB	15.57 s	34.77 m	0.0447	1.6 GB	32.90 s	51.00 m	0.1098
GOD	0.9 GB	10.06 s	15.26 m	0.0474	1.2 GB	21.00 s	17.85 m	0.1218

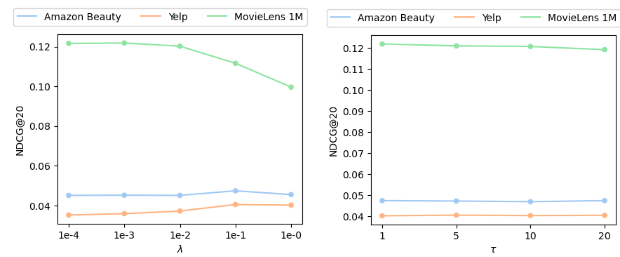


Figure 12: Parameter sensitivity with SASRec.

5.5 Efficiency and Parameter Sensitivity (RQ4)

Beyond performance, practical KD requires efficient training and stable hyperparameter behavior. We therefore analyze the training cost of GOD and its sensitivity to hyperparameters.

Efficiency. Table 7 compares the training efficiency of GOD with representative baselines. Although GOD incurs higher per-epoch time than Student due to hybrid source models and GE/GCL computation, it remains more efficient than BD and EMKD in both memory usage and time per epoch. More importantly, by evaluating student embeddings and encoders through teacher-conditioned source models, GOD provides more direct and fine-grained supervision than output-level imitation. This stronger supervision helps GOD reach its best validation performance in less wall-clock time than BD and EMKD, and even Student, despite the additional per-epoch computation. At the same time, GOD achieves the best NDCG@20 among compact models. These results show that GOD improves recommendation performance without prohibitive training overhead.

Parameter Sensitivity. We analyze the sensitivity of GOD to the loss coefficient λ for $\mathcal{L}_{kd}$ and the temperature τ in $\mathcal{L}_{con}$. As shown in Figure 12, GOD is generally stable across a wide range of τ, indicating that its contrastive distillation is not overly sensitive to temperature selection. The effect of λ is also moderate on Amazon Beauty and Yelp, while MovieLens 1M shows some degradation when λ becomes too large. This may suggest that excessive distillation strength can over-regularize the student on denser datasets, where recommendation supervision is already relatively sufficient. Overall, GOD maintains robust performance under broad hyperparameter ranges, supporting its practical usability.

6 Related Work

6.1 Knowledge Distillation

KD has been widely studied for recommendation to transfer knowledge from a high-capacity teacher to a compact student [2, 24, 57]. Existing methods mainly differ in the form of supervision they distill. Output-level approaches transfer teacher predictions or ranking preferences. For example, RD [47] distills teacher ranking lists, RRD [15] relaxes ranking orders, and CD [23] selects informative items based on teacher rankings for collaborative supervision. Representation-level approaches move beyond final rankings. DE [15] transfers latent embedding knowledge, while HTD [16] distills hierarchical topology in the representation space. Beyond these fixed-teacher methods, BD [21] performs bidirectional distillation by updating the teacher-side knowledge source together with the student. Together, these methods cover output-level, representation-level, and dynamic supervision, but still transfer teacher-produced signals after separate teacher-student execution rather than evaluating student components inside teacher-side computation.

In SR, KD must further transfer dynamic preference patterns from user interaction sequences, making sequence-aware supervision essential. Recent SR-specific methods address this challenge from different perspectives [9, 43, 61]. AdaRec [2] searches for scene-adaptive student architectures and distills knowledge from sequential teachers. MSKDIK [9] distills interest representation and drift knowledge across multiple stages to capture evolving user intent. EMKD [8] leverages ensemble modeling with contrastive and logit-level distillation to provide richer sequential supervision. These studies show that KD for SR benefits from modeling sequential dynamics beyond generic recommendation signals. Nevertheless, they still distill teacher-produced outputs or representations as external targets, leaving student components outside teacher-side sequential computation.

Recently, another line of work leverages large language models (LLMs) for KD, aiming to transfer semantic understanding or reasoning ability to recommenders [52, 58, 65]. For example, DLLM2Rec [5] distills LLM-derived content knowledge into ID-based sequential recommenders, while SLMRec [61] and SLIM [54] distill LLM knowledge or reasoning processes for recommendation. Although these approaches enrich recommendation with external semantic knowledge, they rely on massive LLM teachers, and some distilled students remain language models that are still much larger than ID-based sequential recommenders. This leads to substantial distillation cost and can still incur higher inference latency than ID-based SR models. In contrast, GOD focuses on KD between ID-based sequential recommenders. Rather than injecting external semantic knowledge, GOD improves the generalization of distilled knowledge by providing component-level feedback through grafted teacher-student computation.

6.2 Grafting

Grafting was originally introduced as the inverse operation of pruning in decision trees [56] and later adapted to neural networks to replace or augment selected components. For example, layer grafting [33] improves under-informative layers with validated external layers, NetGraft [41] progressively replaces teacher layers with student counterparts for efficient few-shot KD, and GrafT [35] attaches add-on modules to vision transformers for multi-scale representation sharing. These studies show that grafting can effectively modify selected components without redesigning the entire architecture. Unlike prior grafting methods that mainly target model adaptation or component improvement, GOD uses grafting to construct teacher-conditioned source models for SR. By replacing recommender-specific components, namely embedding tables and sequence encoders, GOD evaluates student components inside teacher-side sequential computation and provides component-level distillation feedback.

7 Conclusion

In this paper, we revisit KD for SR from the perspective of component-level feedback. We identify a limitation of conventional distillation, where teacher-produced signals supervise the complete student path and can entangle the effects of student embeddings and encoders. To address this, we propose GOD, a component-level KD framework that constructs teacher-conditioned source models through grafting. By replacing selected frozen-teacher components with trainable student counterparts, GOD evaluates student components under teacher-side sequential computation and improves the generalization of distilled knowledge. We further introduce GE and GCL to stabilize grafted representation generation and adaptively balance correlated source views. Extensive experiments demonstrate that GOD consistently outperforms existing KD and self-supervised baselines, while maintaining practical training efficiency and using only the non-grafted student for inference. Future work includes extending grafting to heterogeneous teacher-student architectures and exploring more flexible component-level knowledge transfer strategies.

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A GenAI Usage Disclosure

ChatGPT was used solely for English grammar and language refinement. No generative AI was involved in the research process, including but not limited to coding, experiments, and data analysis. All technical contributions are the original work of authors, and AI-assisted edits were manually reviewed to ensure alignment with the original intent.

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